Static Trees

CS3014 - Estructura de Datos Avanzados

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Overview

1.

Goals

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Warming up!

Problem I - Latin American Regional ACM ICPC 2017

https://maratona.sbc.org.br/hist/2017/resultados/contest.pdf

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Do it yourself! (homework) `https://judge.beecrowd.com/en/problems/view/2703`

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- Explore different structures and analyze their performance.

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- Explore different structures and analyze their performance.
- Be able to reduce the LCA problem to RMQ and solve it.
- Solve the Level Ancestor (LA) problem.

Goals

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Problem Scenario	Preprocessing	Space	Query Time
Range Minimum Query (RMQ)	$O(n)$	$O(n)$	$O(1)$
Lowest Common Ancestor (LCA)	$O(n)$	$O(n)$	$O(1)$
Level Ancestor (LA)	$O(n)$	$O(n)$	$O(1)$

Goal: Achieve optimal linear preprocessing and constant time queries across all structures.

2.

RMQ

RMQ

Problem

Given an array A of n numbers (to preprocess). In a query, the goal is to find the minimum element in a range spanned by $A[i]$ and $A[j]$:

$$RMQ(i, j) = \arg\min\{A[i], A[i+1], ..., A[j]\} = k$$

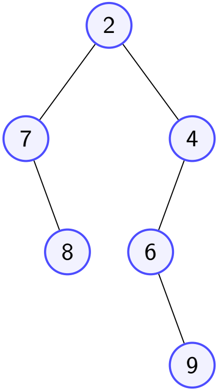
where $i \leq k \leq j$ and $A[k]$ is minimized

3.

RMQ & LCA

Cartesian trees

$A = [7, 8, 2, 6, 9, 4]$



Rules:

- Root = Minimum element $A[m]$
- Left / Right subtrees built recursively.
- Preserves **min-heap property**.

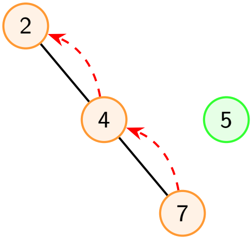
Identity

$\mathrm{RMQ}_A(i, j) \equiv \mathrm{LCA}_T(i, j)$

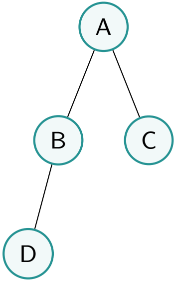
Cartesian trees

Linear time algorithm

- Scan array left to right.
- Walk up the **right spine** until finding $A[v] < A[i]$.
- Insert $A[i]$ as right child; old subtree becomes left child of $A[i]$.
- **Amortized Cost:** Nodes enter/leave spine ≤ 1 time $\implies$ O(n).



LCA to ± 1 RMQ: Euler Tour



Euler Tour (E):	A	B	D	B	A	C	A
Depth Array (D):	**0**	**1**	**2**	**1**	**0**	**1**	**0**

The ± 1 Property

Consecutive depth steps change by exactly $+1$ or -1:

$$|D[i] - D[i-1]| = 1$$

$\text{LCA}(u, v) =$
min value in D between indices of u and v.

4.

Solving RMQ

Solving RMQ

Block partition size: $b = \frac{1}{2}\log_2 n$.

1. Macro-Structure (Across Blocks)

- Sparse Table over block minimums.
- Space: $O(\frac{n}{b}\log\frac{n}{b}) = \mathrm{O}(\mathrm{n})$.

2. Micro-Structure (Inside Blocks)

- Max unique shapes $= 2^{b-1} = \mathrm{O}(\sqrt{\mathrm{n}})$.
- Precompute $b \times b$ lookup tables.
- Space: $O(\sqrt{n} \cdot b^2) = \mathrm{o}(\mathrm{n})$.

Total Performance: O(n) Space / O(1) Query Time

5. Level Ancestor

Level Ancestor

Strategy / Name	**Preprocessing**	**Query Time**	**Core Technique**
Full Table Lookup	$O(n^2)$	$O(1)$	Complete DP lookup table
Jump Pointers	$O(n \log n)$	$O(\log n)$	Binary lifting (powers of 2)
Longest Path Decomposition	$O(n)$	$O(\sqrt{n})$	Disjoint root-to-leaf path arrays
Ladder Decomposition	$O(n)$	$O(\log n)$	Overlapping path extensions (doubled length)
Jump + Ladder	$O(n \log n)$	$O(1)$	1 Jump pointer step + 1 Ladder step
Macro-Micro Tree (Dietz)	$O(n)$	$O(1)$	Leaf trimming & Method of Four Russians

6.

Related

Related

Dynamic

Segment tree, Fenwick tree, ...

7.

Summary

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Thanks

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References

Acknowledgements

The course slides are based on the lectures from previous editions and on similar lectures elsewhere.
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